

Physics-Informed Neural Networks for Formula 1 Tire Degradation

A Parametric Solution Operator with Latent Thermal and Wear States

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September 2026 · Code and data: github.com/SoullessSapo/f1-tire-pinn

Abstract

We present a parametric physics-informed neural network (PINN) that predicts the lap-time loss caused by tire degradation in Formula 1. The model couples two dimensionless ordinary differential equations (ODEs), a lumped thermal balance and an Archard–Arrhenius wear law, whose positive feedback makes the performance *cliff* emerge from the dynamics rather than being hard-coded. Instead of solving one trajectory, the network learns the solution operator over the whole space of race conditions, so predicting a new stint is a single forward pass (0.42 ms on a CPU). The unknown physical coefficients are estimated jointly with the network weights, after identifying and closing two exact scale degeneracies of the inverse problem. On a synthetic bench with known ground truth the PINN reaches an RMSE of 0.063 s, against 0.070 s for an LSTM and 0.246 s for an extended linear model, and recovers nine physical coefficients with a mean error of 1.3%. On public telemetry from eleven races of the 2026 season (434 stints, 8,197 laps), after estimating the fuel-and-track-evolution effect per race and collapsing noisy context proxies to race medians, the PINN beats both baselines on every metric: RMSE 0.694 s versus 0.746 s (LSTM) and 0.801 s (linear), with 5.7% monotonicity violations when extrapolating, versus 11.2% and 26.0%. We also report negative results: on two races of 2023 the PINN loses to the baselines, the original cliff detector was measuring timing noise, and one season of data cannot resolve whether the 2026 compound hierarchy inverted.

Keywords: physics-informed neural networks, inverse problems, tire wear, Formula 1, DeepXDE, motorsport strategy

1. Introduction

Tire degradation drives pit-stop strategy in Formula 1: calling a stop one lap late at the cliff, the point where grip collapses, can cost several positions. Teams usually model degradation with empirical linear fits in seconds per lap, and the machine-learning literature with black-box recurrent networks. Neither knows that a tire cannot regain grip, or that it cannot consume more tread than it has. Both therefore fail exactly where a strategist needs them, when extrapolating beyond the laps observed so far.

Physics-informed neural networks [1] embed differential equations in the training loss, so the network must satisfy the physics even where there is no data. This paper applies that idea to tire wear using only public information: onboard telemetry and lap timing obtained through FastF1 [4]. Our contributions are:

1. A two-ODE thermo-mechanical tire model whose cliff emerges from a positive thermal–wear feedback, together with a pace-loss observable that links the latent states to lap times.
2. A *parametric* PINN that learns the solution operator once and predicts any stint in one forward pass, with initial conditions imposed exactly through an output transform.
3. A joint estimation of the physical coefficients, with an explicit analysis of two exact scale degeneracies that otherwise make the inverse problem diverge while the training loss stays low.
4. An evaluation on synthetic and real data (2023 and 2026) that measures physical consistency alongside accuracy, cross-checks against published degradation figures, and reports negative results as such.

The cloud layer needed for real-time deployment is out of scope; this work covers the physical system, the network, offline training, the baselines and the evaluation.

2. Related Work

Physics-informed learning. Raissi, Perdikaris and Karniadakis [1] formulated PINNs as networks whose loss includes the residual of a differential equation, evaluated by automatic differentiation at collocation points. DeepXDE [2] provides a general implementation, including inverse problems with trainable coefficients; it is the basis of this work, on a PyTorch backend.

Wear and heat. Archard’s law [3] relates worn volume to normal load and sliding distance. We combine it with an Arrhenius-type thermal activation and a lumped-capacitance heat balance of the tread surface.

Degradation in Formula 1. Public analyses summarise degradation as a linear rate in seconds per lap per compound and circuit, estimated from fuel-corrected lap times [5]. Recurrent networks such as LSTMs fit well inside the data but carry no constraint outside it. We use both approaches as baselines.

3. Physical Model

The life of a stint is described in dimensionless time $\tau = \text{stint lap} / L_{\text{ref}}$ by two coupled ODEs with two latent states that are never observed: the thermal excess of the tread surface $\theta = (T_{\text{surface}} - T_{\text{track}}) / \Delta T_{\text{ref}}$ and the fraction of tread consumed $d \in [0, 1]$:

$$\frac{d\theta}{d\tau} = A_{\text{gen}} q (1 + \zeta d) - (h_0 + h_1 v) \theta, \quad (1)$$

$$\frac{dd}{d\tau} = k_w \lambda^m \exp(E_a(\theta + T_{\text{trk}}) - \kappa c) (1 - d). \quad (2)$$

The context variables are the specific frictional energy per lap q , the mean mechanical load λ in g, the mean speed v (which drives convective cooling), the normalised track temperature T_{trk} , and the compound hardness c (0 soft, 1 hard). Equation (1) is a lumped heat balance: frictional generation minus baseline and speed-driven cooling. Equation (2) combines Archard’s law with an Arrhenius activation, so wear grows exponentially with temperature. Two factors do the heavy lifting:

- $(1 - d)$ in (2) bounds $d \in [0, 1]$ *structurally*: the equation itself, not a penalty, prevents consuming more tread than exists.
- $(1 + \zeta d)$ in (1) produces the cliff. As the tread thins, the same frictional energy is deposited into less rubber, the temperature rises, Arrhenius accelerates wear, and the tread thins faster. This positive feedback lets the cliff emerge from the coupled dynamics.

The measurable quantity is not d but the pace loss relative to the best lap of the stint, with $p = 8$:

$$\delta(\tau) = \gamma_1 d + \gamma_2 d^p. \quad (3)$$

The term $\gamma_1 d$ is gradual degradation; $\gamma_2 d^p$ is negligible until d approaches 1 and then dominates, which is the grip collapse.

4. Parametric PINN

4.1 Solution operator

A textbook PINN solves a single trajectory, which would mean retraining for every stint. Here the network is a solution operator

$$\mathcal{N}(\tau, q, \lambda, v, T_{\text{trk}}, c) \mapsto (\theta, d) \quad (4)$$

that learns the whole family of solutions of (1)–(2) across the admissible range of race conditions. It is trained once; predicting a stint is a forward pass with no retraining and no numerical integration. The network is a fully connected MLP with four hidden layers of 64 units, tanh activations and Glorot-normal initialisation (Figure 1).

Initial conditions are imposed exactly through an output transform instead of loss terms, which removes two terms and the weight-balancing problem that most often breaks PINN training:

$$\theta(\tau) = \theta_0 + \tau N_0(x), \quad d(\tau) = \tau \text{softplus}(N_1(x)), \quad (5)$$

so that $\theta(0) = \theta_0$ and $d(0) = 0$ hold exactly and $d \geq 0$ always.

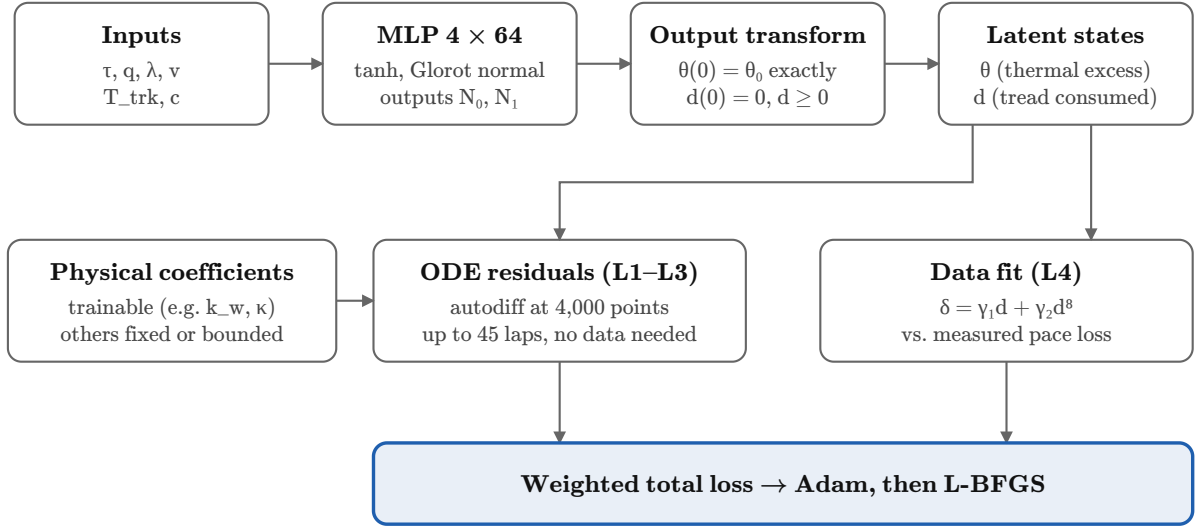


Figure 1. Architecture of the parametric PINN. The ODE residuals are evaluated over the whole hypercube of conditions, including where no lap was observed; the data term only where laps exist. The physical coefficients are optimised jointly with the network weights.

4.2 Loss function

The loss has three physics terms and two data terms (Table 1). The physics terms are enforced at 4,000 uniform collocation points plus dense anchors in τ for every observed context, out to the full 45-lap decision horizon rather than only as far as the longest observed stint. That is the advantage over a black box: outside its training distribution an LSTM has nothing tying it to thermodynamics.

Table 1. Terms of the loss function and their weights. Weights in parentheses are those used on real data.

Term	Enforces	Evaluated on	Weight
L1	residual of (1)	whole condition hypercube	1
L2	residual of (2)	whole condition hypercube	1
L3	bound $d \leq d_{\max} = 1.1$	whole condition hypercube	10
L4	fit to measured pace loss	observed laps only	20 (5)
L5	temperature proxy (synthetic only)	observed laps only	1 (off)

Training uses Adam (learning rate 10^{-3}) followed by L-BFGS. The second-order phase is not optional: the thermal coefficients ζ , h_0 and h_1 reach the pace loss only through two layers of composition, stall throughout the Adam phase, and converge only once L-BFGS takes over (Figure 4). On real data the weight of the data term is lowered from 20 to 5, because a real lap carries about 0.5 s of noise and at the full weight its plateau would dominate the ODE terms indefinitely.

4.3 Inverse problem and its degeneracies

The coefficients $\zeta, h_0, h_1, k_w, m, E_a, \kappa, \gamma_1, \gamma_2$ are unknown and are estimated jointly with the network weights as trainable variables. ODE coefficients are parametrised in log space, so they are positive by construction; bounded coefficients use a sigmoid into their declared range. The model has two exactly degenerate directions, and ignoring them does not produce a mediocre fit but divergence:

1. **Thermal scale.** Doubling A_{gen} and halving E_a leaves wear unchanged. This is closed by fixing A_{gen} , which anchors the temperature scale.
2. **Wear scale.** The transformation $d \rightarrow \varepsilon d$, $\gamma_1 \rightarrow \gamma_1/\varepsilon$, $\gamma_2 \rightarrow \gamma_2/\varepsilon^p$ leaves δ exactly unchanged. On synthetic data the thermal proxy and the stints that saturate at $d = 1$ break it; real data has neither. In one run on Monza and Hungary 2023 the optimiser slid along this direction to $\gamma_2 = 2.5 \times 10^{13}$ and a test RMSE of 5.8×10^9 s, while the training loss stayed low (0.117). It is closed by bounding $\gamma_1 \in [0.2, 4.0]$ s and $\gamma_2 \in [0.2, 6.0]$ s, which asserts something genuinely known: a destroyed tire costs a few seconds a lap, not millions.

The general lesson is that in a PINN inverse problem a low training loss guarantees nothing if the model has degenerate directions; they must be enumerated and closed explicitly. On real data only k_w and κ are left free, the two quantities that genuinely vary between circuits and compounds, while the rest are pinned at calibrated values. For 2026, κ switched from a log to a symmetric sigmoid on $[-1.5, 1.5]$, so the sign of the compound effect is decided by the data rather than by the parametrisation.

5. Data

We use a synthetic bench with known ground truth and two sets of real telemetry obtained with FastF1 (Table 2). Train and test sets are always split by whole stint, never by lap, to prevent leakage within a stint.

Table 2. Datasets. Stint length is given as minimum / median / maximum laps; compounds as hard / medium / soft stints.

Dataset	Races	Stints	Laps	Stint length	H / M / S
Synthetic	–	64	1,520	12 / 24 / 34	21 / 21 / 22
Real 2023	Monza, Hungary	36	740	8 / 20 / 35	14 / 21 / 1
Real 2026	11 races, Melbourne to Zandvoort	434	8,197	8 / 18 / 42	225 / 167 / 42

5.1 Synthetic bench

The bench integrates (1)–(2) with known coefficients using fourth-order Runge–Kutta and adds measurement noise ($\sigma = 0.06$ s on pace, 0.05 on θ). The PINN starts from deliberately wrong initial values and must recover the true ones from the observed pace loss alone, which validates the inverse problem in a way real data cannot. Each stint is cut two to five laps *after* the tire passes $d = 0.85$, imitating a strategist who loses laps deciding, waiting for a pit window or covering a rival. That margin matters: cutting exactly at the cliff gave a mean parameter-recovery error of 11.8% (γ_2 off by 62%), and keeping those few laps lowered it to 2.1%.

5.2 Real telemetry and proxies

Nothing the model needs is directly observable: internal temperature, vertical load and tread state are proprietary to each team. We therefore build proxies from public telemetry:

- q , the specific frictional power, from integrating $|a|v$ over the lap (the heat-generation term of (1));
- λ , the mean total acceleration in g (the Archard term of (2)); lateral acceleration is not in the telemetry and is reconstructed by differentiating the GPS trajectory twice after Savitzky–Golay smoothing over a fixed time window;
- v , the mean speed; T_{trk} , the track temperature from session weather; and c , the compound.

Only green-flag, accurate, non-deleted laps without pit entry or exit on fresh sets are kept, since $d(0) = 0$ only holds for a new tire. Proxies are made dimensionless against fixed references rather than each session’s median, which would put every circuit at 1.0 and erase the between-circuit variation the model has to learn.

5.3 Observable and race-lap correction

The observable is the pace loss relative to the stint’s performance *peak*. A new set comes out cold and gets faster for two or three laps; the model is monotone by construction and cannot represent that warm-up, so $d = 0$ is anchored at the peak and earlier laps are discarded. On 2023 data this anchoring lowered the PINN’s RMSE from 2.82 s to 0.57 s and its monotonicity violations from 16.4% to 0%.

A car sheds about 100 kg of fuel in a race and the circuit rubbers in. Both effects are smooth monotone functions of race lap and cannot be separated from each other, but their sum can be estimated per race with

$$t_{\text{lap}} \sim \text{driver} + f(\text{race lap}) + \text{degradation}(\text{tire age, compound}), \quad (6)$$

where f is a piecewise-linear spline. Identification comes from cars carrying tires of different ages at the same race lap (a spread of 2–7 laps, with correlations of 0.22–0.76 between age and race lap). The estimated effect ranges from -0.026 s/lap at Miami to -0.097 s/lap at Spa, against the constant -0.055 assumed previously. Over a 20-lap stint that constant produces a bias between -0.57 s and $+0.83$ s, with different signs at different circuits, which is comparable to the whole degradation signal.

6. Experimental Setup

Baselines. (i) A *linear* model representing the empirical approach used by teams, generously extended with a quadratic term in τ and interactions between τ and the context, fitted by ridge regression so that it is not a straw man. (ii) An *LSTM* with 48 hidden units over the lap sequence of a stint, representing the black-box approach.

Metrics. RMSE, MAE and maximum error of the pace loss on unseen test stints; cliff-lap error; and the *monotonicity violation rate*, the share of lap-to-lap steps where a model predicts the tire regaining grip, measured both inside the observed stint and when extrapolating to 45 laps. The physically correct violation rate is 0%, and no error metric penalises it, so it is measured separately.

Cliff definition. A cliff is a pace-loss slope of at least 0.30 s/lap sustained over four consecutive laps, after a three-lap moving average. Section 7.4 explains why a weaker criterion was abandoned.

Training. Synthetic bench: 64 stints split 48 / 16, 15,000 Adam iterations and 3,000 L-BFGS iterations, about 30 minutes on a CPU. Real 2026 data: 326 / 108 stints, 12,000 Adam and 2,500 L-BFGS iterations. All runs use seed 42 and reproduce exactly. The full configuration is listed in Appendix A.

7. Results

7.1 Synthetic bench

The PINN is the most accurate model and the most physically consistent: it has no monotonicity violations inside the observed range, against 22.7% for the linear model (Table 3). The difference is starkest when extrapolating (Figure 2): the LSTM bends downwards, predicting that the tire regains grip, and the linear model can go negative, while the PINN keeps integrating the ODE and saturates.

Table 3. Test results on the synthetic bench (16 unseen stints). Violations are the share of lap-to-lap steps predicting a gain in grip.

Model	RMSE [s]	MAE [s]	Max. error [s]	Viol. interp.	Viol. extrap.
PINN	0.063	0.050	0.189	0.0%	1.1%
LSTM	0.070	0.057	0.219	0.8%	1.1%
Linear	0.246	0.159	0.913	22.7%	22.2%

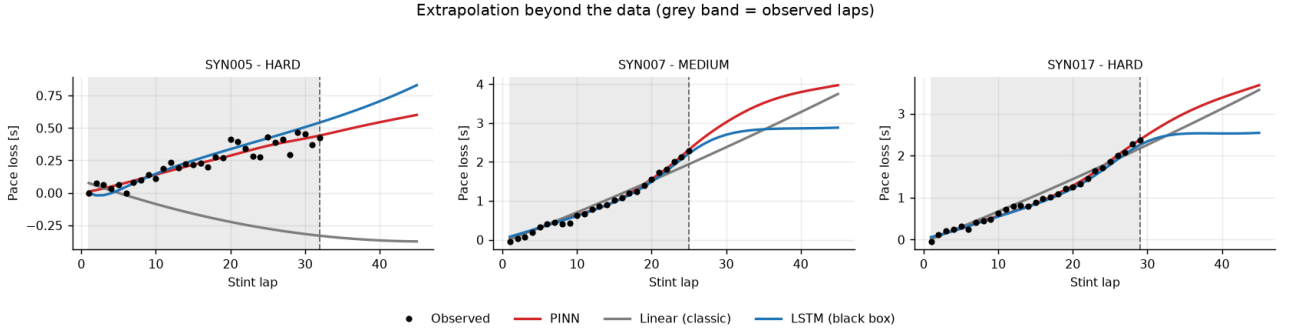


Figure 2. Extrapolation on the synthetic bench. The grey band marks the observed laps; to the right of the dashed line every model is extrapolating. The LSTM (blue) bends towards a gain in grip and the linear model (grey) can drop below zero, while the PINN (red) follows the ODE and saturates.

Latent states. The wear state d never appears in the training data, yet the network reconstructs it closely (Figure 3), purely because it is forced to satisfy the differential equations. A network without physics has no way to recover a state it never observes.

Latent states: d is reconstructed from the ODE alone (never observed); θ carries weak supervision

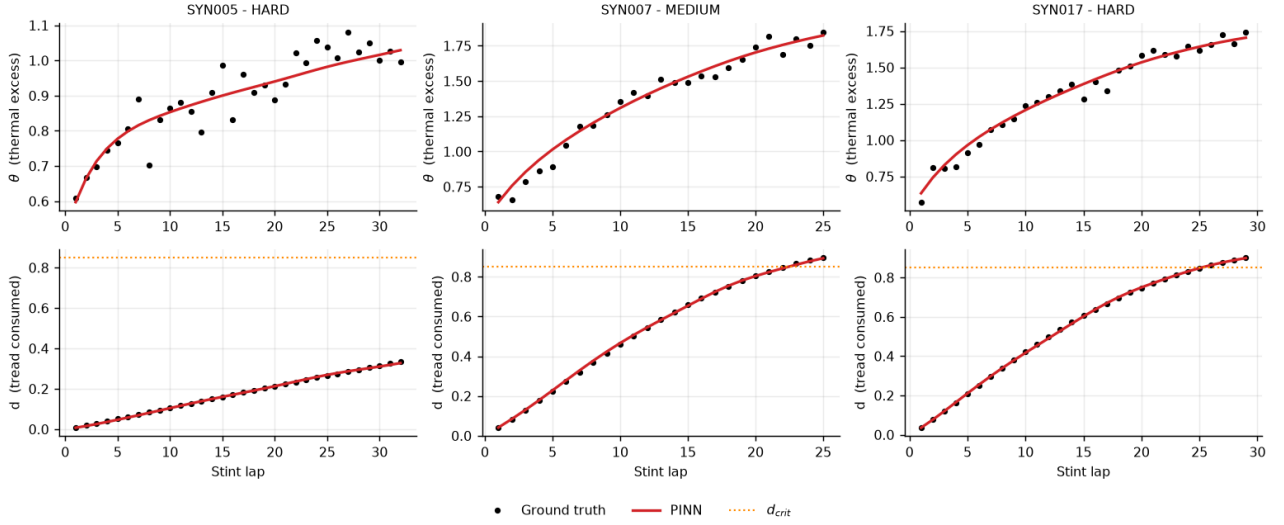


Figure 3. Latent states reconstructed by the PINN on test stints. Top: thermal excess θ , which receives weak supervision. Bottom: tread consumed d , which is never observed and is recovered from the ODE alone. The dotted line marks $d_{\text{crit}} = 0.85$.

Parameter recovery. Starting from deliberately wrong values, the PINN recovers all nine coefficients using only the observed pace loss, with a mean relative error of 1.3% (Table 4, Figure 4).

Table 4. Recovery of the physical coefficients on the synthetic bench.

Coefficient	Meaning	Initial	Estimated	True	Error
ζ	wear-temperature coupling (cliff)	0.50	0.893	0.900	0.7%
h_0	baseline cooling	4.0	5.968	6.000	0.5%
h_1	forced convection	2.0	3.998	4.000	0.0%
k_w	wear rate	0.80	0.565	0.550	2.6%
m	load exponent	1.00	1.494	1.500	0.4%
E_a	thermal activation	0.60	0.930	0.950	2.1%
κ	compound hardness	0.50	0.852	0.850	0.2%
γ_1	linear pace loss [s]	1.00	1.350	1.350	0.0%
γ_2	cliff magnitude [s]	1.50	2.738	2.600	5.3%
Mean relative error					1.3%

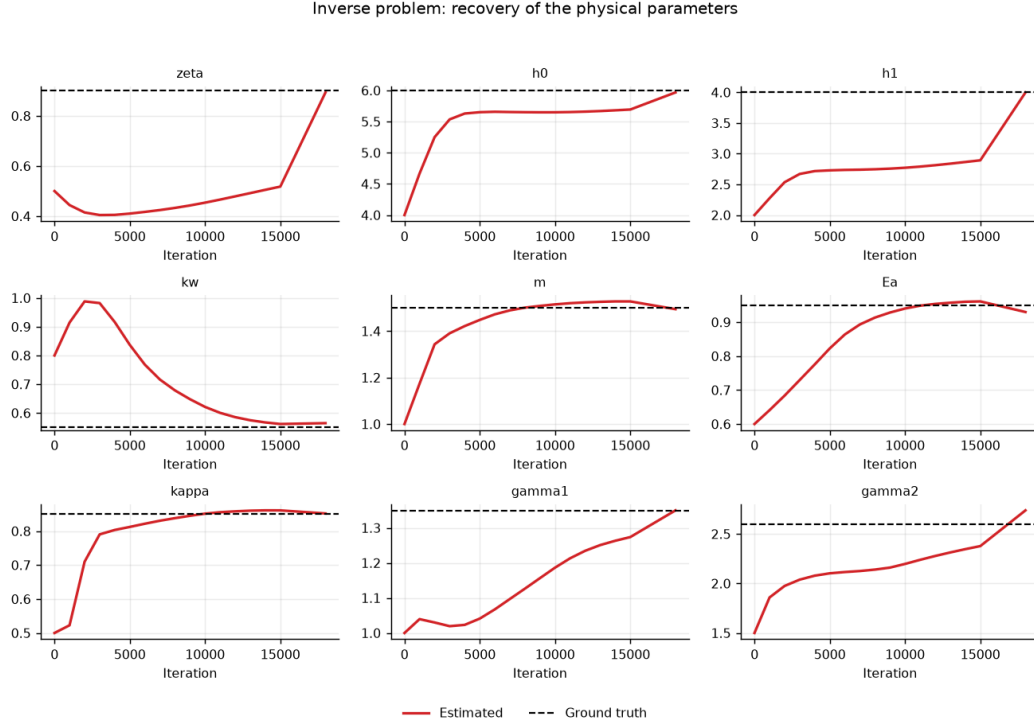


Figure 4. Convergence of the estimated coefficients (red) towards the ground truth (dashed). Note the jump in ζ , h_0 and h_1 after iteration 15,000, when L-BFGS takes over from Adam.

7.2 Real telemetry, 2023 (two races)

On Monza and Hungary 2023 (36 stints) the PINN does *not* beat the baselines (Table 5). With so few stints, a noise floor of about 0.5 s per lap and little variation in conditions, the observed curve is close to linear over the measured range, and a linear fit is hard to beat. The fitted coefficients are physically reasonable ($k_w = 0.813$, $\kappa = 0.968$) and neither sits on a bound, but the wear-ODE residual settles around 0.15, two orders of magnitude worse than on the synthetic bench.

Table 5. Test results on Monza and Hungary 2023.

Model	RMSE [s]	MAE [s]	Max. error [s]	Viol. interp.	Viol. extrap.
LSTM	0.536	0.423	2.023	0.6%	8.8%
Linear	0.539	0.429	1.920	0.0%	0.0%
PINN	1.172	0.725	3.493	6.3%	6.3%

7.3 Real telemetry, 2026 season (eleven races)

The hypothesis that more races would let the physics pay off did not hold at first: with the initial configuration the PINN scored an RMSE of 0.964 s against 0.768 s for the LSTM. Two corrections to the *data*, not to the model, changed the outcome (Table 6).

Per-race race-lap effect. Replacing the fixed fuel constant by the estimate of (6) lowered the PINN’s RMSE to 0.928 s. It moved the baselines more than the PINN: their extrapolation violations roughly tripled, to 29.0% (linear) and 10.1% (LSTM), because the over-correction had been adding a spurious upward trend that masked their tendency to bend downwards.

Context proxies collapsed to race medians. A variance decomposition showed that 53% of the variance of q and 61% of that of λ occurs between stints at the *same* circuit, and that this within-circuit variation predicts nothing: correlated against the within-circuit deviation of measured degradation it gives

$r = 0.001$ and $r = 0.029$, against a 5% critical value of 0.095 ($n = 423$). It is measurement noise from double-differentiated GPS. Between circuits the same proxies do carry signal, and track temperature is the strongest predictor ($r = 0.664$). Collapsing q and λ to their race medians therefore discards noise and keeps signal. Speed is left per stint because its within-circuit deviation is predictive ($r = -0.203$, with the physically expected sign).

Table 6. Test RMSE [s] on the 2026 season as the data pipeline was corrected. The oracle fits a separate straight line to each test stint and marks the noise floor.

Data pipeline	PINN	LSTM	Linear	Oracle
Fixed fuel correction (-0.055 s/lap)	0.964	0.768	0.796	0.538
+ race-lap effect estimated per race	0.928	0.748	0.817	–
+ q , λ collapsed to race medians	0.694	0.746	0.801	0.531

With the final pipeline the PINN beats both baselines on every metric (Table 7), the first time it does so on real data. It is also the only model that detects any of the six cliffs present in the test set (4 of 6, with a mean error of 6.25 laps). The asymmetry in Table 6 is mechanistic: the baselines use the context only as regression features, where noise attenuates a coefficient, whereas the PINN imposes the physics *as a function of* the context at collocation points across the whole hypercube, so noisy context coordinates corrupt the constraint everywhere. The more a model leans on its inputs, the more noise in them costs.

Table 7. Final test results on the 2026 season (108 unseen stints).

Model	RMSE [s]	MAE [s]	Max. error [s]	Cliffs found	Viol. interp.	Viol. extrap.
PINN	0.694	0.494	4.023	4 / 6	2.7%	5.7%
LSTM	0.746	0.541	4.032	0 / 6	9.9%	11.2%
Linear	0.801	0.602	4.499	0 / 6	7.9%	26.0%

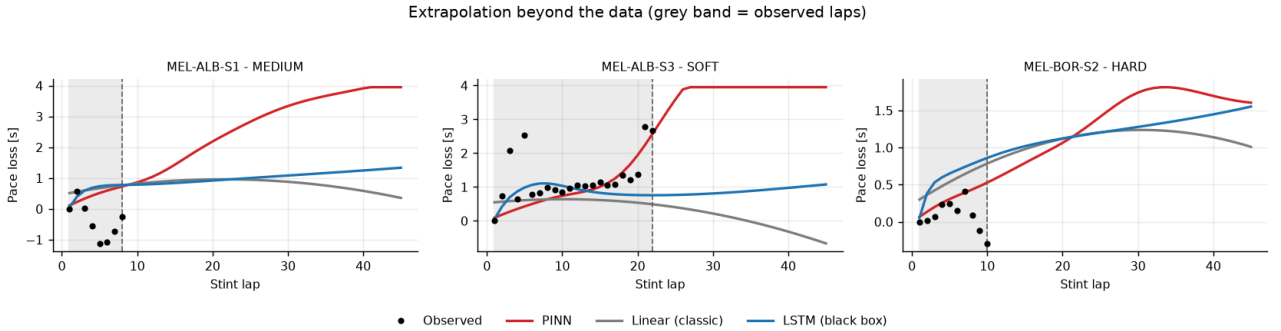


Figure 5. Extrapolation on three 2026 test stints. Real stints are short and noisy; beyond the data the linear model and the LSTM bend towards a gain in grip, while the PINN keeps degrading. The PINN is not free of violations on real data (right panel), which is reflected in its 5.7% extrapolation rate.

Decision map. Because the network is parametric, a full strategy map costs one batched forward pass: Figure 6 shows the lap at which d crosses 0.85 for 1,728 combinations of compound, track temperature and load. Broadly, wear arrives earliest at high load and high track temperature and soonest on the soft compound, but the map is not strictly monotone in either variable, which marks the limit of what eleven races can teach the model about conditions.

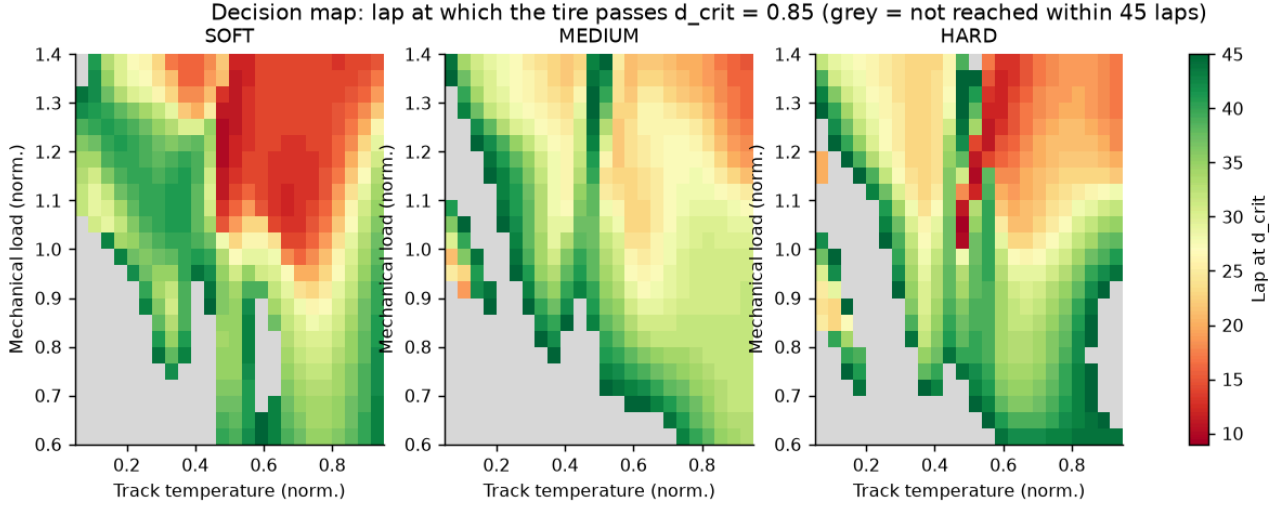


Figure 6. Decision map from the 2026 model: lap at which the tire passes $d_{crit} = 0.85$ by compound, normalised track temperature and mechanical load. Red cells wear out early; grey cells survive the 45-lap horizon.

7.4 Verification against published degradation data

We cross-checked the model against a public analysis of F1 tire degradation [5], measuring degradation the same way: a linear fit of fuel-corrected pace loss against stint lap. The 2023 compound spread (medium minus hard) is 0.0102 s/lap here against 0.011 s/lap published, a difference of about 7%. The model’s predicted rates are close to those observed: 0.0892 versus 0.0916 s/lap for the medium (−2.6%) and 0.0755 versus 0.0814 s/lap for the hard (−7.2%). The synthetic bench is also well calibrated in magnitude without having been tuned for it: a nominal medium stint degrades at 0.086 s/lap against a real median of 0.090 s/lap.

The original cliff detector was measuring noise. The first criterion, a slope above 0.15 s/lap at any single lap, fires on curves that contain no cliff once realistic timing noise is present (Table 8). It explained a result that should have looked suspicious: 34 of 36 real 2023 stints “had a cliff” while their median degradation was a steady 0.09 s/lap. The sustained-slope criterion of Section 6 lowers false positives to 0–1% while still catching 96–100% of genuine knees; under it only 2 of those 36 stints show a cliff. The model predicts *degradation curves* well; “cliff-lap prediction” oversells what race data supports, because as a detectable event the cliff barely occurs in current Formula 1.

Table 8. False-positive rate of the original single-point cliff test on cliff-free curves, by timing noise.

Timing noise σ [s]	0.00	0.05	0.10	0.20	0.30	0.50
False “cliff” detected	0%	0.6%	45%	98%	100%	100%

The 2026 compound hierarchy cannot be resolved. The published analysis reports that in 2026 the hard compound degrades fastest (0.071 s/lap, against 0.065 for the medium and 0.063 for the soft). Controlling for circuit, driver and the race-lap effect, we measure a hard-minus-medium difference of -0.0061 ± 0.0063 s/lap over eleven circuits (95% CI $[-0.018, +0.006]$, $t = -0.96$), with five of eleven circuits going each way. The interval contains the published value, zero and the classic ordering alike: the standard deviation of the effect across circuits (0.021 s/lap) is three times the effect being sought. An earlier version of this analysis reported -0.027 s/lap and a clear classic ordering; that figure came from the biased fixed fuel correction without driver control and is retracted. The fitted $\kappa = +0.031$ likewise indicates essentially no compound effect in this data.

7.5 Inference latency

Predicting a full 45-lap stint takes 0.42 ms on average and 1.47 ms at the 95th percentile on a CPU (500 repetitions). A 500 ms end-to-end budget for live strategy is therefore consumed by transport, not by the model.

8. Discussion and Limitations

The PINN’s advantage does not come from extra capacity but from constraints that hold outside the data, so it only shows when there is enough data and the context is clean: it loses with little variation (2023) and wins with eleven circuits and denoised proxies (2026). Two findings generalise beyond tires. First, an inverse problem with a low training loss can still be diverging along a degenerate direction. Second, a model that imposes physics as a function of its inputs is more sensitive to input noise than a regression that uses the same inputs as features.

The main limitations are:

- **Constant context per stint.** q , λ and v are summarised by a stint or race median; lap-to-lap variation is absorbed by the data residual.
- **λ is a relative proxy.** It comes from double-differentiated GPS and its absolute magnitude (about 3–4 g) exceeds what an accelerometer would read, though it is monotone in true load and discriminates between circuits.
- **Fuel burn and track evolution** are only estimated together; the data cannot separate them.
- **Traffic and dirty air** are mitigated by filters, not removed.
- **γ_2 is weakly identified.** It acts only as $d \rightarrow 1$ and teams pit before that, so the information about the final collapse is largely absent from race data. Relatedly, k_w and γ_1 partially compensate each other; only the saturation of long stints pins the scale of d .
- **Internal temperature is not public**, so the thermal proxy is disabled on real data and more coefficients must be fixed at calibrated values.

Natural continuations are several seasons of data to resolve compound effects, load proxies that do not rely on twice-differentiated GPS, context that varies within a stint (fuel load, traffic) as τ -dependent inputs, and the real-time cloud deployment left out of scope here.

9. Conclusion

A parametric PINN built on two coupled ODEs predicts Formula 1 tire degradation more accurately and more consistently with physics than an extended linear model and an LSTM, both on a synthetic bench (RMSE 0.063 s, coefficient recovery within 1.3%) and on a full season of real telemetry (RMSE 0.694 s against a noise floor of 0.531 s), at sub-millisecond inference cost. The main lessons are methodological: close the degeneracies of the inverse problem explicitly, anchor degradation at the performance peak, estimate the race-lap effect per circuit rather than assume it, and remove noise from the context, which hurts a model more the more it relies on its inputs.

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Appendix A. Configuration

Table 9. Hyperparameters. Values in the right column differ for the real-data runs.

Group	Setting	Synthetic	Real (2026)
Scales	L_{ref} (laps per unit τ), ΔT_{ref}	30, 40 K	same
	θ_0 , d_{max} , A_{gen} (fixed), p	0.5, 1.1, 8, 8	same
Network	hidden layers, activation	4×64 , tanh	same
	collocation points, τ anchors per context	4,000, 40	same
Training	Adam iterations ($\text{lr } 10^{-3}$)	15,000	12,000
	L-BFGS iterations	3,000	2,500
	weights L1 / L2 / L3 / L4 / L5	1 / 1 / 10 / 20 / 1	1 / 1 / 10 / 5 / –
	free coefficients	all nine	k_w , κ
Bounds	γ_1 , γ_2 , κ	[0.2, 4.0] s, [0.2, 6.0] s, [–1.5, 1.5]	
Strategy	d_{crit} , horizon	0.85, 45 laps	
	cliff slope, minimum run	0.30 s/lap, 4 laps	
Data	references q , λ , v	–	1,900 W/kg, 3.8 g, 58 m/s
	min. stint laps, max. pace loss kept, test fraction	12–34 laps, –, 0.25	8 laps, 6.0 s, 0.25